

Practical: Customer Information System using HBase and HiveQL

Aim

To implement a **Customer Information System** using **HiveQL** and **HBase** which includes:

1. Creating Customer, Order, and Item tables in Hive
 2. Loading data from local storage
 3. Performing JOIN operations in Hive
 4. Creating Index on tables
 5. Finding Total and Average Sales
 6. Finding Order Details with Maximum Cost
 7. Connecting Hive with HBase using External Table
 8. Displaying Customer Information records from HBase
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Software Requirements

- Hadoop
 - Hive
 - HBase
 - Linux/Ubuntu
 - Java
-

Step 1: Start Hadoop and HBase Services

Start Hadoop

```
start-dfs.sh  
start-yarn.sh
```

Start HBase

```
start-hbase.sh
```

Start Hive

hive

Step 2: Create Hive Database

```
CREATE DATABASE customerdb;
```

Use Database

```
USE customerdb;
```

Step 3: Create Tables in Hive

A. Create Customer_Info Table

```
CREATE TABLE customer_info(  
    cust_id INT,  
    cust_name STRING,  
    order_id INT  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```

B. Create Order_Info Table

```
CREATE TABLE order_info(  
    order_id INT,  
    item_id INT,  
    quantity INT  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```

C. Create Item_Info Table

```
CREATE TABLE item_info(  
    item_id INT,  
    item_name STRING,  
    item_price INT  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```

Step 4: Create Data Files in Local Storage

customer.txt

```
101,Rahul,5001  
102,Priya,5002  
103,Amit,5003  
104,Neha,5004
```

order.txt

```
5001,201,2  
5002,202,1  
5003,203,4  
5004,201,3
```

item.txt

```
201,Laptop,50000  
202,Mobile,20000  
203,Headphone,2000
```

Step 5: Load Data into Hive Tables

```
LOAD DATA LOCAL INPATH '/home/hadoop/customer.txt'  
INTO TABLE customer_info;
```

```
LOAD DATA LOCAL INPATH '/home/hadoop/order.txt'  
INTO TABLE order_info;
```

```
LOAD DATA LOCAL INPATH '/home/hadoop/item.txt'  
INTO TABLE item_info;
```

Step 6: Display Table Records

Customer Table

```
SELECT * FROM customer_info;
```

Order Table

```
SELECT * FROM order_info;
```

Item Table

```
SELECT * FROM item_info;
```

Step 7: Perform JOIN Operation in Hive

Join Customer, Order, and Item Tables

```
SELECT  
c.cust_id,  
c.cust_name,  
o.order_id,  
i.item_name,  
i.item_price,  
o.quantity,  
(i.item_price * o.quantity) AS total_cost  
FROM customer_info c  
JOIN order_info o  
ON c.order_id = o.order_id  
JOIN item_info i  
ON o.item_id = i.item_id;
```

Expected Output

cust_id	cust_name	order_id	item_name	item_price	quantity	total_cost
101	Rahul	5001	Laptop	50000	2	100000
102	Priya	5002	Mobile	20000	1	20000
103	Amit	5003	Headphone	2000	4	8000
104	Neha	5004	Laptop	50000	3	150000

Step 8: Create Index on Customer Information Table

Create Index

```
CREATE INDEX idx_customer  
ON TABLE customer_info(cust_name)  
AS 'COMPACT'  
WITH DEFERRED REBUILD;
```

Rebuild Index

```
ALTER INDEX idx_customer  
ON customer_info  
REBUILD;
```

Show Indexes

```
SHOW INDEXES ON customer_info;
```

Step 9: Find Total Sales in Hive

```
SELECT SUM(i.item_price * o.quantity) AS total_sales  
FROM order_info o  
JOIN item_info i  
ON o.item_id = i.item_id;
```

Expected Output

total_sales

278000

Step 10: Find Average Sales in Hive

```
SELECT AVG(i.item_price * o.quantity) AS average_sales
FROM order_info o
JOIN item_info i
ON o.item_id = i.item_id;
```

Expected Output

average_sales

69500

Step 11: Find Order Details with Maximum Cost

```
SELECT
o.order_id,
i.item_name,
o.quantity,
(i.item_price * o.quantity) AS total_cost
FROM order_info o
JOIN item_info i
ON o.item_id = i.item_id
ORDER BY total_cost DESC
LIMIT 1;
```

Expected Output

order_id	item_name	quantity	total_cost
5004	Laptop	3	150000

Step 12: Open HBase Shell

Exit Hive:

```
exit;
```

Open HBase shell:

```
hbase shell
```

Step 13: Create Customer Information Table in HBase

```
create 'customer_hbase',
```

```
'customer_details',
```

```
'order_details'
```

Step 14: Insert Records into HBase Table

```
put 'customer_hbase','101','customer_details:name','Rahul'
```

```
put 'customer_hbase','101','order_details:order_id','5001'
```

```
put 'customer_hbase','102','customer_details:name','Priya'
```

```
put 'customer_hbase','102','order_details:order_id','5002'
```

```
put 'customer_hbase','103','customer_details:name','Amit'
```

```
put 'customer_hbase','103','order_details:order_id','5003'
```

```
put 'customer_hbase','104','customer_details:name','Neha'
```

```
put 'customer_hbase','104','order_details:order_id','5004'
```

Step 15: Display Records in HBase

Scan Complete Table

```
scan 'customer_hbase'
```

Expected Output

ROW COLUMN+CELL

```
101  customer_details:name = Rahul
```

```
101  order_details:order_id = 5001
```

```
102  customer_details:name = Priya
```

```
102  order_details:order_id = 5002
```

Step 16: Create External Hive Table Connected to HBase

Exit HBase shell:

```
exit
```

Open Hive:

```
hive
```

Use Database

```
USE customerdb;
```

Create External Hive Table

```
CREATE EXTERNAL TABLE customer_hive(  
    cust_id STRING,  
    cust_name STRING,  
    order_id STRING  
)  
STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'  
WITH SERDEPROPERTIES (  
    "hbase.columns.mapping" =  
    ":key,
```



```
customer_details:name,  
order_details:order_id"  
)  
TBLPROPERTIES ("hbase.table.name" = "customer_hbase");
```

Step 17: Display External Hive Table Data

```
SELECT * FROM customer_hive;
```

Expected Output

cust_id	cust_name	order_id
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101	Rahul	5001
102	Priya	5002
103	Amit	5003
104	Neha	5004

Viva Questions

1. What is Hive?

Hive is a data warehouse system used for querying and managing large datasets using HiveQL.

2. What is HBase?

HBase is a distributed NoSQL database built on top of Hadoop HDFS.

3. What is HiveQL?

HiveQL is an SQL-like language used in Apache Hive.

4. What is an External Table?

An external table maps external data without storing data inside Hive warehouse.

5. Why use HBase with Hive?

- HBase provides real-time data access
 - Hive provides SQL querying support
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Conclusion

Successfully implemented a Customer Information System using Hive and HBase by:

- Creating Hive tables
- Loading local data
- Performing joins
- Calculating sales statistics
- Creating indexes
- Connecting Hive with HBase
- Displaying HBase records through Hive external table